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Obstacle Avoidance Formation Strategy for Unmanned Vehicles via Improved Grey Wolf Optimizer and Artificial Potential Field Method

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Abstract: This paper proposes an obstacle avoidance formation control strategy for multi-unmanned vehicles, integrating an improved grey wolf optimizer (IGWO) with a kinematics-integrated artificial potential field (APF) method. By embedding steering constraints and velocity limits into a kinematic-coupled APF model, path feasibility is ensured by adaptive attraction-repulsion forces. Furthermore, the IGWO enhances parameter optimization efficiency by employing a nonlinear convergence factor and a multi-objective fitness function that simultaneously optimizes path length, formation stability, and safety constraints. To improve adaptability in complex environments, a dynamic formation-switched strategy is introduced, which autonomously adjusts formation patterns and APF gains based on real-time obstacle density and channel width.

Key Words: Multi-unmanned vehicles; Artificial potential field; Grey wolf optimizer; Obstacle avoidance; Formation control.

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1 Introduction

With the rapid development of intelligent technologies, multi-agent systems in security patrols, logistics scheduling, and vehicular networks face escalating operational complexity. The dynamic balance between environmental avoidance and formation coordination poses a critical challenge in multidimensional spatial tasks, demanding robust algorithms for real-time adaptability and optimization.

The Grey Wolf Optimizer (GWO) was first proposed by Mirjalili. [1], simulating the hierarchical leadership and cooperative hunting behaviors of grey wolves. This algorithm provided a novel approach for solving complex optimization problems in engineering domains. Building on this foundation, [2] applied GWO to optimize control systems in power grids, addressing stability challenges under dynamic load variations and communication constraints. An enhanced Artificial Potential Field method was developed in [3], combining gravitational attraction and repulsive fields to improve obstacle avoidance and path tracking in autonomous vehicles. In [4] further improved APF by reconstructing potential fields to resolve local minima issues, enhancing path planning reliability in constrained environments. Researchers in [5] designed a GWO-based stabilizer for power systems, focusing on mitigating communication delays and maintaining stability in wide-area networks. Authors in [6] introduced a binary variant of GWO for feature selection tasks, balancing global exploration and local exploitation through adaptive parameter updates. A hybrid algorithm combining GWO with Differential Evolution was proposed in [7] for UAV path planning, effectively generating collision-free trajectories in complex environments. Authors in [8] enhanced GWO by integrating opposition-based learning strategies, improving global search capabilities and convergence performance. Extending GWO to multi-objective optimization, [9] introduced an external archive mechanism to manage Pareto-optimal solutions, ensuring diversity in multi-criterion problems. In [10] developed a mutation-guided GWO variant for feature selection, leveraging correlation analysis to maintain population diversity and avoid premature convergence. In control system design, authors in [11] utilized GWO to optimize fuzzy controller parameters, reducing sensitivity to system uncertainties and nonlinearities. In [12] proposed a rotated APF method combined with formation control models, enabling stable 3D coordination for multi-UAV systems. Recent advancements in [13] integrated velocity potential fields with probabilistic filters for dynamic obstacle detection and real-time path adaptation. In [14] reconstructed APF frameworks using behavioral models to enhance obstacle avoidance strategies in autonomous driving scenarios. Finally, [15] comprehensively analyzed motion planning frameworks for autonomous vehicles, bridging theoretical principles with practical implementation challenges in urban environments.

Compared to existing approaches, this study introduces the following contributions:

1) Kinematics-Coupled Potential Field Model: In contrast to [12] and [14], which neglect kinematic constraints in path planning, our model embeds steering angle and velocity limitations directly into the artificial potential field formulation. This ensures dynamically feasible paths that strictly adhere to the agents' motion capabilities, resolving infeasible trajectories caused by conventional geometric-only constraints.

2) Enhanced GWO with Nonlinear Convergence Factors: Unlike the traditional GWO in [1] and its multi-objective variants in [9], which suffer from premature convergence in late iterations, we propose a nonlinear convergence factor adjustment mechanism. This improvement accelerates optimization efficiency during the exploitation phase. Furthermore, the enhanced GWO dynamically tunes the gain coefficients of the potential field, addressing the static parameter limitations

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of APF methods.

3) Adaptive Formation Switching and Holistic Fitness Evaluation: Compared to [7] and [15], we develop an adaptive formation switching strategy that selects optimal patterns based on real-time obstacle density and passage width. Concurrently, a unified fitness function integrates path length, formation stability, and task completion time, enabling comprehensive performance quantification for multi-agent systems—a capability absent in prior works.

2 System Model and Problem Formulation

The unmanned vehicle adopts a differential drive structure with the kinematic equations as follows:

$$\begin{cases} V_x = V_l\cos(\theta) + V_r\cos(\theta), \\ V_y = V_l\sin(\theta) - V_r\sin(\theta), \\ \dot{\theta} = \frac{V_r - V_l}{L_s} \end{cases} \tag{1}$$

where V_x and V_y means the velocity components of the unmanned vehicle in the x and y directions, respectively; L_s is the wheelbase; V_l and V_r means the linear velocities of the left and right wheels; θ is the heading angle and $\dot{\theta}$ means the rate of θ. Assume that the current position of the unmanned vehicle is $q = (x_q, y_q)$, and the target position is $q_g = (x_{qg}, y_{qg})$. Then the heading angle and its angular deviation relative to the target direction is given as follows:

$$\begin{aligned} \vec{d} &= (x_{qg} - x_q, y_{qg} - y_q) \\ \theta &= \arctan 2(V_y, V_x) \\ \theta_g &= \arctan 2(y_{qg} - y_q, x_{qg} - x_q) \\ \phi &= |\theta - \theta_g| \end{aligned} \tag{2}$$

where $\vec{d}$ is target direction vector; θ_g is target direction angle; ϕ denotes included angle. The steering constraints and velocity limits considered in this paper is shown as follows:

$$|V_l|, |V_r| \le V_{\max} \text{ and } \theta_{\max} = \arctan\left(\frac{L_s\omega_{\max}}{2V_{\max}}\right) \tag{3}$$

where $\omega_{\max}$ is the maximum angular velocity. Considering the kinematic constraints of the considered unmanned vehicle, we reconstruct an attraction-repulsion force model based on an improved APF method. The improved gravitational field function is given as follows:

$$U_{\text{att}}(q) = \frac{1}{2}k_{\text{att}}\|q - q_g\|^2\left(1 + \alpha e^{-\beta\|q-q_g\|}\right)\left[1 - \frac{\phi}{2\theta_{\max}}\right]$$

where k_{att} is the attractive gain; α and β are parameters that regulate the intensity and speed of gravitational decay. And the improved repulsive field function is given as follows:

$$U_{\text{rep}}(q) = \begin{cases} \frac{1}{2}k_{\text{rep}}\left(\frac{1}{d_{\text{eff}}} - \frac{1}{d_0}\right)^2\left(1 + \frac{\vec{v}_o\cdot(q-q_o)}{\|\vec{v}_o\|d_{\text{eff}}}\right), & d_{\text{eff}} \le d_0 \\ 0, & d_{\text{eff}} > d_0 \end{cases}$$

where k_{rep} is the repulsive force gain; q_o is the obstacle position and d_{eff} is the effective obstacle avoidance distance, satisfying $d_{\text{eff}} = \|q - q_0\| - \frac{w}{2} - r_0$; d_0 is the influence range threshold of the repulsive potential field; v_0 is the relative velocity of the obstacle.

Then, the attraction vector $\vec{F}_{\text{att}}$ and repulsion vector $\vec{F}_{\text{rep}}$ are yielded by solving the negative gradient of the potential field function:

$$\vec{F}_{\text{att}} = -k_{\text{att}}\vec{r}\left[\left(1 + \alpha e^{-\beta r}\right) - \frac{\alpha\beta r}{2}e^{-\beta r}\right]\left[1 - \frac{|\phi|}{2\theta_{\max}}\right]$$

$$\vec{F}_{\text{rep}} = \begin{cases} k_{\text{rep}}\left(\frac{1}{d_{\text{eff}}} - \frac{1}{d_0}\right)\left[\frac{1}{d_{\text{eff}}^2} + \frac{\vec{v}_o\cdot(q-q_o)}{\|\vec{v}_o\|d_{\text{eff}}^3}\right] \\ \times\left(1 + \frac{\vec{v}_o\cdot(q-q_o)}{\|\vec{v}_o\|d_{\text{eff}}}\right)\frac{q-q_o}{\|q-q_o\|}, & d_{\text{eff}} \le d_0 \\ 0, & d_{\text{eff}} > d_0 \end{cases}$$

where $r = \|q - q_g\|$. Finally, the resultant force $\vec{F}_{\text{total}}$ is generated by considering the influence of all obstacles as follows:

$$\vec{F}_{\text{total}} = \vec{F}_{\text{att}} + \sum_{k=1}^{N_{\text{obs}}}\vec{F}_{\text{rep},k} \tag{4}$$

where $\vec{F}_{\text{rep},k}$ represents the repulsion from the k-th obstacle and N_{obs} denotes the number of obstacles per unit area.

Problem Formulation: The main objective of this paper is to design an obstacle avoidance formation strategy for unmanned vehicles, integrating an IGWO with a kinematics-integrated APF method.

3 Main Results

3.1 Improved grey wolf optimizer

In APF method, the attractive and repulsive gain coefficients, which are core parameters directly determining obstacle avoidance performance, are typically manually configured based on empirical knowledge. This static parameterization approach often fails to adapt to the optimization demands of dynamic environments, leading to issues such as local minima entrapment and path redundancy. This section proposes an IGWO to achieve adaptive optimization of potential field gain coefficients, thereby enhancing convergence efficiency in obstacle avoidance and improving path planning quality.

First, we will show the traditional GWO, in order to highlight the improved GWO proposed in this paper. The individual's position information consists of the attractive gain coefficient k_{att} and the repulsive gain coefficient k_{rep}. The individual's position update formula is given as follows:

$$k(m+1) = k_p(m) - AD$$

where $k(m)$ denotes the position of a wolf at the m-th iteration and $k_p(m)$ is the position of the current optimal solution; A determines the search strategy; D represents the distance between individual wolves and the prey. These parameters are calculated as:

$$A = 2ar_1 - a, \quad D = |Ck_p(m) - k(m)|, \quad C = 2r_2$$

where C defines the search range of the wolves; a is the convergence factor that linearly decreases from 2 to 0 as iterations progress, balancing global exploration and local exploitation. Here, r_1 and r_2 are random numbers uniformly distributed in $[0, 1]$, introducing stochasticity.

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What should be mentioned is that, the convergence factor a in the traditional GWO decreases linearly with increasing iterations, leading to excessively slow convergence in later stages. In this paper, the following nonlinear convergence factor is designed to improve that in the traditional GWO:

$$a(m) = 2 - 2\left(\frac{m}{M}\right)^{\eta}, \quad \eta > 1 \tag{5}$$

where m is the current iteration index and M is the maximum number of iterations. The curve of the nonlinear convergence factor $a(m)$ is shown in Figure 1.

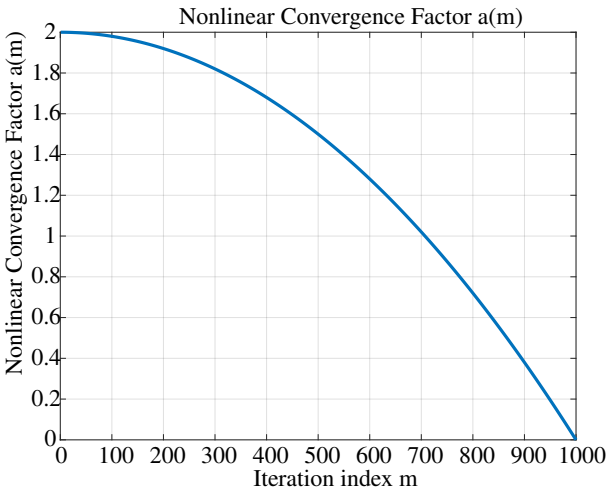


Fig. 1: Nonlinear convergence factor a(m)

The fitness function evaluates the overall performance of formation path planning by integrating multiple metrics, such as path length, formation stability, steering angle constraints, and inter-vehicle safety distance. By adjusting the weight coefficients ω_1, ω_2, ω_3, and ω_4, the relative importance of these objectives is balanced to determine the optimal path planning solution. In this paper, the fitness function is designed as follows:

$$\begin{aligned}\mathcal{F} = \omega_1 \frac{L}{L_{\text{opt}}} + \omega_2 \sigma_{\text{form}} + \omega_3 \sum_{i=1}^{N}\sum_{t=1}^{T} \max\left(0, |\theta_{i,t}| - \theta_{\max}\right) \\ + \omega_4 \sum_{i=1}^{N}\sum_{j\neq i} \max\left(0, d_{\text{safe}} - d_{ij}\right)\end{aligned} \tag{6}$$

where L is the total path length of all unmanned vehicles in the formation and L_{opt} is the theoretical shortest path length; σ_{form} is the formation stability metric; $\theta_{i,t}$ is the steering angle of the i-th unmanned vehicle at time t and $\theta_{\max}$ is the maximum allowed steering angle; d_{ij} is the effective distance between the i-th and j-th unmanned vehicles, satisfying $\|q_i - q_j\| - \frac{w}{2}$; w is the vehicle width; d_{safe} is the minimum safe distance between unmanned vehicles.

In GWO, the optimization problem is typically formulated as a minimization task. Therefore, solutions with smaller fitness values are considered superior. The objective of the fitness function is to find its minimum value Minimize $\mathcal{F}$. In this paper, the historical best solution $k^* = \arg\min \mathcal{F}$ is introduced to guide the search process, enabling the search agent to better utilize the excellent solutions found.The update formula for the individual's position is as follows:

$$k(m+1) = k(m) + \lambda(k^*(m) - k(m)) + AD \tag{7}$$

where $\lambda \in (0.5, 1)$ is a dynamic weight, shifting from group exploration to elite guidance.

$$\left|\frac{\text{Minimize}\mathcal{F}(m+1) - \text{Minimize}\mathcal{F}(m)}{\text{Minimize}\mathcal{F}(m+1)}\right| < \epsilon \text{ or } m > M_{\max}$$

where $M_{\max}$ is the maximum number of iterations and ϵ is the preset threshold with $\epsilon = 0.01$. Finally, the resultant force in the APF method is calculated using the optimal gain coefficients, and it is applied to control the position of the leader in the leader-follower approach.

3.2 Improved Leader-Follower Formation Strategy

During the formation transformation process, it is necessary to precisely control the position of each intelligent body to ensure the accuracy and stability of the formation. Based on the leader-follower model, this paper establishes the relative relationship control method between the leader's position q_l and the follower's position q_f:

$$V = \min\left(\|\vec{F}_{\text{total}}\| \cdot V_{\max}, V_{\max}\right)$$

$$\theta = \text{saturate}\left(\arctan\left(\frac{\omega L_s}{V}\right), \theta_{\max}\right)$$

$$q_l(t+\Delta t) = q_l(t) + \begin{bmatrix} V\cos(\theta) \\ V\sin(\theta) \end{bmatrix} \Delta t \tag{8}$$

$$q_f = q_l + \delta R(\vartheta) S p_{\text{ref}} \tag{9}$$

where ω is the leader's angular velocity; Δt is the control cycle; $q_l(t)$ is the current position of the leader at the t-th iteration, while $q_l(t+\Delta t)$ is the position of the leader at the end of a control cycle. ϑ is the entire formation rotation angle, $R(\vartheta)$ is the rotation matrix used to adjust the direction of the follower relative to the leader, enabling the team to rotate based on the leader's motion state or environmental requirements, their calculation formulas are as follows:

$$y' = -(x_0 - x_l)\sin\theta + (y_0 - y_l)\cos\theta$$

$$d_{forward} = V\Delta t$$

$$\vartheta = \arctan\left(\frac{y'}{d_{forward}}\right)$$

$$R(\vartheta) = \begin{bmatrix} \cos\vartheta & -\sin\vartheta \\ \sin\vartheta & \cos\vartheta \end{bmatrix}$$

where y' represents the lateral offset distance of the obstacle relative to the leader, $q_0 = (x_0, y_0)$ is the obstacle position, $q_l = (x_l, y_l)$ is the leader position, and $d_{forward}$ is the forward distance traveled by the leader; $S = \text{diag}(s_x, s_y)$ is the scaling matrix used for formation scaling, satisfying $s_x \cdot s_y \geq \left(\frac{d_{\text{safe}}}{d_{\text{avg}}}\right)^2$; s_x and s_y are the scaling factors in the x and y directions, respectively;δ is defined as a dynamic adjustment factor based on the minimum obstacle distance $d_{\min}$ and the minimum safe distance d_{safe} within the team $\delta = 1/(1+e^{-k(d_{\min} - d_{\text{safe}})})$, when the minimum distance $d_{\min}$ exceeds the safe distance d_{safe}, the value of δ approaches 1,

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which results in a tighter formation with smaller relative distances between agents. when $d_{\min}$ approaches or falls below the safe distance d_{safe}, the value of δ decreases, leading to a looser formation, which helps avoid potential collisions; k is the adjustment coefficient controlling the change rate of the dynamic adjustment factor; p_{ref} is the reference position of the follower relative to the leader, defining the initial shape of the team. The leader-follower formation model is shown in Figure 2, where F_{total} is the total force vector acting on the leader.

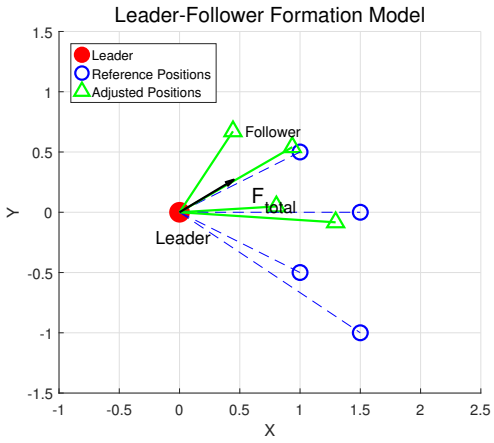


Fig. 2: Leader-Follower Formation Model

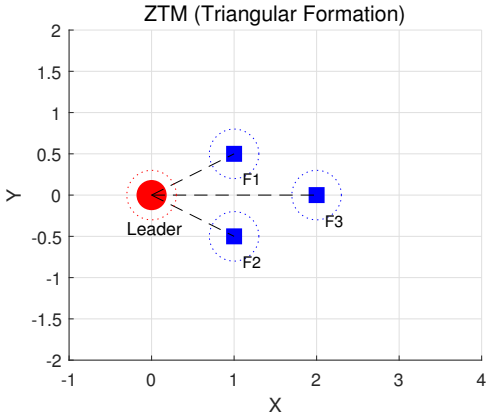


Fig. 3: Zero Transformation Mode

3.3 Dynamic Formation Switched Strategy

The formation mode switching is triggered by environmental perception (obstacle density $\rho_{\text{obs}} = \frac{N_{\text{obs}}}{\pi R^2}$, channel width W) and coupled with the adjustment of potential field gain coefficients to achieve self-adaptive optimization.

Zero Transformation Mode (ZTM): As shown in Figure 3. When $\rho_{\text{obs}} < 0.3$ and $W > 3R_{\min}$, a triangular formation is maintained. Then we reduce k_{rep} to allow a looser formation shape and increase k_{att} to accelerate movement towards the target.

Homogeneous Transformation Mode (HTM): As depicted in Figure 4. When $0.3 \leq \rho_{\text{obs}} < 0.6$ or $2R_{min} \leq W \leq 3R_{\min}$, the formation size is shrinked proportionally. Then we increase k_{att} to enforce a tighter formation shape and increase k_{rep} to enhance obstacle avoidance capability.

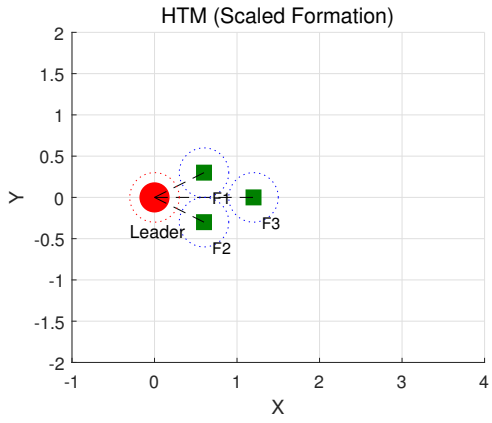


Fig. 4: Homogeneous Transformation Mode

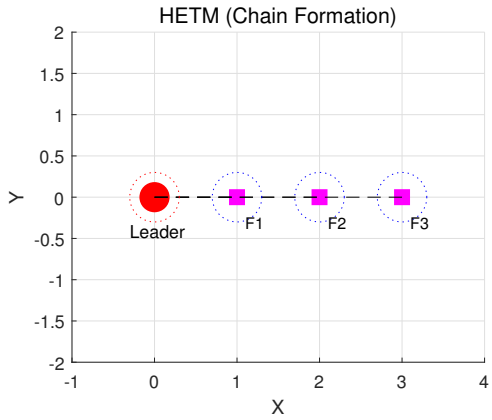


Fig. 5: Heterogeneous Transformation Mode

Heterogeneous Transformation Mode (HETM): As illustrated in Figure 5. When $\rho_{\text{obs}} \geq 0.6$ or $W < 2R_{\min}$, the formation is switched to a chain one. Then we increase k_{rep} to prevent internal collisions within the chain formation, decrease k_{att} to avoid excessive tightness of the formation.

Here, N_{obs} is the number of obstacles per unit area, R is the radius range considering the influence of obstacles and $R_{\min}$ is the minimum safety radius.

Through the above formation strategy, the position of the follower can be dynamically adjusted based on the leader's motion state and environmental information, achieving the dynamic transformation and maintenance of the formation. Based on the entire process described in this paper, we propose the following algorithm framework:

4 Conclusion

This paper proposes an obstacle avoidance method integrating adaptive artificial potential fields with optimization algorithms, addressing the challenges of low path feasibility, inefficient parameter optimization, and poor adaptability to dynamic obstacles in mobile robot formations.A kinematic-coupled potential field ensures feasible paths, while an enhanced Grey Wolf Optimizer boosts parameter search efficiency. A formation-switching strategy dynamically balances obstacle avoidance and formation integrity.

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Algorithm 1 Multi-mode Formation Strategy with IGWO and APF

Step 1. Initialize parameters:
- Set kinematic constraints: $V_{\max}$, $\theta_{\max}$, L_s
- Set the initial parameters: k_{att}, k_{rep}, a_{init}, $T_{\max}$, ϵ
- Define formation modes: ZTM, HTM, HETM

Step 2. Build kinematic-coupled APF:
- Calculate attraction field:
  $U_{\text{att}}(q) = \frac{1}{2}k_{\text{att}}\|q-q_g\|^2\left(1+\alpha e^{-\beta\|q-q_g\|}\right)\left[1-\frac{\phi}{2\theta_{\max}}\right]$
- Calculate repulsion field:
  $U_{\text{rep}} = \frac{1}{2}k_{\text{rep}}\left(\frac{1}{d_{\text{eff}}}-\frac{1}{d_0}\right)^2\left(1+\frac{\vec{v}_o\cdot(q-q_o)}{\|\vec{v}_o\|d_{\text{eff}}}\right)$
- Total force: $\vec{F}_{\text{total}} = \vec{F}_{\text{att}} + \sum_{k=1}^{N_{\text{obs}}}\vec{F}_{\text{rep},k}$

Step 3. IGWO-based parameter optimization:
- Update nonlinear convergence factor:
  $a(m) = 2-2\left(\frac{m}{M}\right)^{\eta}, \quad \eta > 1$
- Evaluate fitness for each wolf using (6).
- Update the gain coefficients using (7).

Step 4. Dynamic formation switching:
- **If** $\rho_{\text{obs}} < 0.3$ & $W > 3R_{\min}$:
  Activate ZTM mode: $k_{\text{att}} \uparrow, k_{\text{rep}} \downarrow$
- **Else if** $\rho_{\text{obs}} \geq 0.6 \parallel W \leq 3R_{\min}$:
  Activate HTM/HETM mode: Scale formation via s_x, s_y; adjust $k_{\text{att}}, k_{\text{rep}}$

Step 5. Formation control execution:
- Update leader positions:
  $q_l(t+\Delta t) = q_l(t) + \begin{bmatrix} V\cos(\theta) \\ V\sin(\theta) \end{bmatrix}\Delta t$
- Follower position adjustment:
  $q_f = q_l + \delta R(\vartheta)Sp_{\text{ref}}$

Step 6. Iterate until convergence:
- **If** $\left|\frac{\text{Minimize}\mathcal{F}(m+1)-\text{Minimize}\mathcal{F}(m)}{\text{Minimize}\mathcal{F}(m+1)}\right| < \epsilon$ or $m > M_{\max}$:
  Output optimal path and formation parameters.
- **Else**:
  Repeat **Steps 3-5** with updated environmental data.

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